



BUSINESS
SCHOOL

SKEMA

Financial Analysis and Decision Making

Global BBA Fall 2026

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Session 4: Rates



About me

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Excel Functions



- ▷ `-PV(rate,nper,pmt)` & `-FV(rate,nper,pmt)`
 - Gives the PV or FV of an annuity

- ▷ `=PMT(rate,nper,pv)`
 - Solves for C in the annuity formula

- ▷ `=NPV(rate,value1,[value2],...)`
 - Assumes $[value1]$ is C_1

Excel Functions



	A	B	C	D	E	F	G
1	C_0	C_1	C_2	C_3	C_4	C_5	C_6
2	-100	50	50	50	50	50	50
3	Discount Rate	10%					
4	NPV	83					
5	IRR	45%					

Formula in cell B4: = A2 + NPV (B3, B2 : G2)

Formula in cell B5: = IRR (A2 : G2)

Excel Functions



	A	B	C	D	E	F	G
1	C_0	C_1	C_2	C_3	C_4	C_5	C_6
2	-100	50	50	50	50	50	50
3	Discount Rate	10%					
4	NPV	83					
5	IRR	45%					

Alternatively in cell B4: = A2 - PV(B3, COLUMNS(B2:G2), B2)

where PV(rate,nper,pmt)

Excel Functions



	A	B	C	D	E	F	G
1	C_0	C_1	C_2	C_3	C_4	C_5	C_6
2	-100	?	?	?	?	?	?
3	Discount Rate	10%					
4	NPV	83					
5	IRR	45%					

What if we want to know C (assuming $C_1, \dots, C_6$ are equal)?

= - PMT (B3, COLUMNS (B2 : G2) , B4-A2)

Example



Suppose the present value of an apartment you would like to purchase is \$500,000. You plan a 30-year mortgage with equal annual payments, at an interest rate of 5% compounded annually. What is your yearly payment?

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$$PV = \frac{C}{r} \left[1 - \left(\frac{1}{1+r} \right)^T \right]$$

$$PV \times r = C \times \left[1 - \left(\frac{1}{1+r} \right)^T \right]$$

$$\frac{PV \times r}{\left[1 - \left(\frac{1}{1+r} \right)^T \right]} = C$$

Example



Suppose the present value of an apartment you would like to purchase is \$500,000. You plan a 30-year mortgage with equal annual payments, at an interest rate of 5% compounded annually. What is your yearly payment?

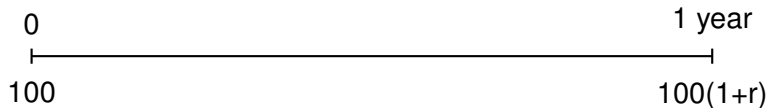
$$C = \frac{PV \times r}{1 - (1 + r)^{-T}}$$
$$C = \frac{500 \times 0.05}{1 - (1.05)^{-30}} = 32.526$$

Effective Annual Rate



Suppose a bank offers you .001% interest compounded every 6 hours. Is this a lot? A little? As a means of standardization, the **effective annual rate (EAR)** gives you the amount of interest that will be earned at the end of one year.

Annual Compounding



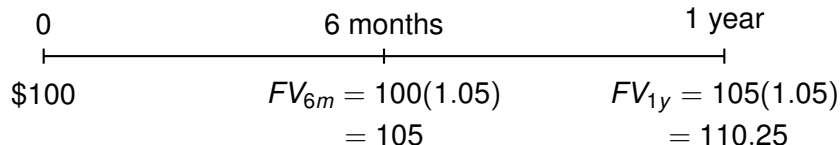
So far, whenever we solved problems using a given r , we assumed *annual compounding*.

Example. How much money would you have in your bank account if you deposit €100 in your bank account at an interest rate of 10%?

Semi-Annual Compounding



What if your bank offered you an annual interest rate of 10%, compounded semi-annually?



You made **10.25%** after 1 year, NOT 10%.

We call 10.25% the **effective annual rate**.

Quoting Rates



“Suppose a bank offers you a semi-annual rate of 10%”

≠

“Suppose a bank offers you an annual rate of 10%, compounded semi-annually”

≠

“Suppose a bank offers you an effective annual rate of 10%”

Quoting Rates



In general, the standard way of quoting rates is

$r\%$ annually, compounded m times a year

such that you earn

$$\text{Deposit} \times \left(1 + \frac{r}{m}\right)^m$$

where compounded

- ▷ daily: $m = 365$
- ▷ monthly: $m = 12$
- ▷ quarterly: $m = 4$
- ▷ annually: $m = 1$

Quoting Rates



In general, the standard way of quoting rates is

$r\%$ annually, compounded m times a year

such that you earn

$$\text{Deposit} \times \left(1 + \frac{r}{m}\right)^m$$

and

- ▷ r : the nominal rate, or the annual percentage rate (APR)
- ▷ $\frac{r}{m}$: the proportional rate

Effective Annual Rate



We want

$$\text{Deposit} \times \left(1 + \frac{r}{m}\right)^m = \text{Deposit} \times (1 + \text{EAR})$$

Exercise 5.4



Compute the EAR for each investment choice

- a. 9% APR compounded monthly (9.38%)
- b. 12% APR compounded annually (12%)
- c. 9% APR compounded daily (9.416%)

Exercise 5.6



Your bank account pays interest with an EAR of 9%. What is the APR quote for this account based on semiannual compounding? What is the APR with monthly compounding?

(8.806%,8.649%)

Equivalent Discount Rate



The effective annual rate, as the name implies, is the total amount of interest you receive after one year.

Define the **equivalent n -period discount rate** as the total amount of interest you receive after time n .

* For all time-value of money calculations, use the EAR or the n -period discount rate.

Equivalent Discount Rate



The EAR and the n -period equivalent discount rate is related in the following way

$$(1 + \text{EAR})^n = 1 + r_{\{n\}}$$

where $r_{\{n\}}$ denotes the n -period equivalent discount rate.

Example. Suppose the EAR is 10%. What is the equivalent 6-month discount rate? (2.47%)

Equivalent Discount Rate



Example. Suppose the equivalent 2-year discount rate 15%. What is the effective annual rate?

(7.238%)

Strategy for Solving Questions



1. Convert the interest rate to an effective annual rate
2. Find the appropriate n -period discount rate using

$$(1 + \text{EAR})^n = 1 + r_{\{n\}}$$

- If necessary, use the relevant n -period discount rate to discount cash flows that arrive every n -periods

Exercise 5.1



Your bank is offering you an account that will pay 10% interest in total for a two-year deposit. Determine the equivalent discount rate for a period length of:

- a. Six months

(EAR = 4.88% and equivalent 6-month discount rate = 2.41%)

Exercise 5.1



Your bank is offering you an account that will pay 10% interest in total for a two-year deposit. Determine the equivalent discount rate for a period length of:

- b. One year
- c. One month
(0.397%)

Exercise



Suppose the equivalent 6-month discount rate is 4%. What is the equivalent 2-year discount rate?

(EAR = 8.16% and equivalent 2-year discount rate = 16.985%)

Exercise



Suppose a bank offers you an APR of 10% compounded semi-annually. What is the equivalent 1-month discount rate?
(EAR = 10.25% and equivalent 1-month discount rate = 0.816%)

Exercise



You can buy the car for \$65,000 up front, or you can lease it for \$1,800 per month for 36 months. The firm can borrow at an interest rate of 8% APR with quarterly compounding. Should you purchase the system outright or pay \$1,800 per month?

(equivalent 1-month discount rate = 0.006627% and $PV=57,486$)

Exercise 5.2



Which do you prefer: a bank account that pays 8% per year (EAR) for three years or:

- a. An account that pays 4% every six months for three years?
(.2653 vs .2597)
- b. An account that pays 0.8% per month for three years?
(.3322 vs .2597)